

QUESTION OF THE WEEK (WEEK 4)

CLASS XI (sub: PHYSICS)

- 1) a) What are the limitations of dimensional analysis?
 b) Check the correctness of the physical relation $\rho = \frac{3g}{4\pi r g}$ where symbols carry their usual meanings.
 c) Find the value of 20 J on the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units.
- 2) a) State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ .
 b) Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force.
- 3) a) Define the coefficient linear, superficial and cubical expansions. Derive relations between them.
 b) Why a thick glass cracks when boiling water is poured in it?
- 4) a) Pendulum clocks generally go fast in winter and slow in summer. Why?
 b) A pendulum clock loses 12 Sec. a day if the temperature is 40°C and gains 4 Sec. a day if the temperature is 20°C . Find the temperature at which the clock will show correct time and coefficient of linear expansion of the metal of pendulum shaft.
- 5) a) State and explain the principle of superposition of Coulomb's force.
 b) Two-point charges of $10\ \mu\text{C}$ and $-40\ \mu\text{C}$ are separated by distance of 13 cm. Find the point along the line joining them at which net electrostatic force on any charge is zero.
- 6) a) Define dielectric constant. Why it's value is always greater than one?
 b) Four equal charges q are placed at the four corners of a square of length a . Find the magnitude of the net force acting on one of them due to other charges and draw diagram to show the direction of force on that charge.
- 7) a) Explain why vector addition is commutative, providing a real-life example to support your reasoning.
 b) What is meant by resolution of vectors? How will you resolve a vector in a plane in two mutually perpendicular directions?
- 8) a) Why does a solid expand on heating?
 b) Explain an experiment to determine the coefficient of linear expansion of a solid.
 c) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder at 15°C is 8.0 cm and there is an all-round clearance of 0.05 mm between the piston and the cylinder of the wall. Find the temperature at which the two will exactly fit in. ($\alpha_{\text{steel}} = 1.2 \times 10^{-5}\ ^\circ\text{C}^{-1}$ and $\alpha_{\text{aluminium}} = 1.6 \times 10^{-5}\ ^\circ\text{C}^{-1}$)
- 9) a) State and explain Coulomb's law in electrostatics.
 b) Why coulomb's law is applicable to stationary charges only?
 c) Define 1 coulomb charge on the basis of Coulomb's law.
 d) Two charges 10^{-6}C and 10^{-7}C attract each other with a force of 400N. Calculate the distance between them if they are in i) air and ii) water (relative permittivity of water is 80)